

PCOSINA THESIS APPENDIX: AUTH, ONBOARDING, AND DATA INTEGRITY

****Scope****

This appendix documents the end-to-end auth/onboarding/data flow analysis, formal behavioral specifications, and engineering

A) Diagnosis (Root Causes + Locations)

1. Validation gaps in auth

- Root cause: Email format and password complexity not enforced before Firebase calls.
- Components: AuthViewModel.onLogin, AuthViewModel.onSignUp.
- Files: app/src/main/java/com/pcosina/app/ui/AuthViewModel.kt.

2. Onboarding resets after relogin

- Root cause: Routing decisions made before profile load completes; stale or default profile used.
- Components: AppNavHost login route; UserViewModel load timing.
- Files: app/src/main/java/com/pcosina/app/ui/navigation/AppNavHost.kt.

3. Cross-account plan and grocery leakage

- Root cause: Email-keyed DataStore and in-memory caches not cleared on logout.
- Components: UserPreferencesRepository keys; MealPlanViewModel; GroceryViewModel.
- Files: app/src/main/java/com/pcosina/app/data/repository/UserPreferencesRepository.kt, app/src/main/java/com/pcosina/app/data/repository/GroceryRepository.kt.

4. Identity drift risk

- Root cause: Email is mutable; storage keyed by email breaks when email changes.
- Components: UserPreferencesRepository key strategy.
- Files: app/src/main/java/com/pcosina/app/data/repository/UserPreferencesRepository.kt.

B) Correct Behavior Spec (Formal Rules)

1. Signup/Login validation

- Email required and valid format.
- Password required, length ≥ 8 , at least 1 letter, 1 number, 1 special character.
- Confirm password must match.
- Invalid input must not call Firebase; show inline error.

2. onboardingComplete/profileCompleted persistence

- Profile completion is per-account and stored by UID.
- Routing must wait for profile load before deciding destination.

3. Logout behavior

- Must sign out Firebase.
- Must clear session and reset in-memory caches.
- Must not delete user data unless explicitly requested.

4. Per-account isolation

- All persisted data is UID-keyed.
- No read/write when UID missing.
- Reset in-memory caches on UID change.

C) Fix Plan (Ordered, Minimal Risk)

1. Enforce auth validation rules in AuthViewModel.

2. Reset UserViewModel, MealPlanViewModel, GroceryViewModel on logout.

3. Route to Splash after login and wait for profile load to decide Dashboard vs Onboarding.

4. Switch DataStore keys from email to UID.

5. Migrate legacy email-keyed data to UID once.

6. Invalidate in-memory caches on UID change.

D) Improvements & Enhancements

Demo polish (high-impact, low-effort)

1. Inline field validation and helpful hints for onboarding.

2. Numeric input masks for age, weight, height, budget.

3. Disable Next until required fields are valid.

4. Loading overlay during profile restoration after login.

5. Offline error banner for plan generation.

6. Show plan timestamp on plan screen.

7. Explain why each profile field is required.

8. Distinct errors for auth vs network failures.